

第五节 隐函数的求导公式

一、一个方程的情形

二、方程组的情形

一、一个方程的情形

引例:已知 $e^{x+y} + xy = 0$ 确定 $y = y(x)$, 求 $y'(x)$

一般地, $F(x, y) = 0$ 可确定可导函数 $y = y(x)$,
如何求导?

1. $F(x, y) = 0$

隐函数存在定理 1 设函数 $F(x, y)$ 在点 $P(x_0, y_0)$ 的某一邻域内具有连续的偏导数, 且 $F(x_0, y_0) = 0$, $F_y(x_0, y_0) \neq 0$, 则方程 $F(x, y) = 0$ 在点 $P(x_0, y_0)$ 的某一邻域内恒能唯一确定一个单值连续且具有连续导数的函数 $y = f(x)$, 它满足条件 $y_0 = f(x_0)$,

并有

$$\frac{dy}{dx} = -\frac{F_x}{F_y}.$$

隐函数的求导公式

例 1 验证方程 $x^2 + y^2 = 1$ 在点 $(0,1)$ 的某个邻域内能唯一确定一个有连续导数的隐函数 $y = f(x)$, 当 $x = 0$ 时, $y = 1$, 并求 $f'(0), f''(0)$.

解 令 $F(x, y) = x^2 + y^2 - 1$,

则 $F_x = 2x, F_y = 2y, F(0,1) = 0, F_y(0,1) = 2 \neq 0$,

依定理 6.5.1 知方程 $x^2 + y^2 = 1$ 在点 $(0,1)$ 的某个邻域内能唯一确定一个有连续导数的函数 $y = y(x)$, 当 $x = 0$ 时 $y = 1$,

且
$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{x}{y},$$

$$\begin{aligned}
 \frac{d^2 y}{dx^2} &= \frac{d}{dx} \left(-\frac{x}{y} \right) = -\frac{y - xy'}{y^2} \\
 &= -\frac{y - x(-\frac{x}{y})}{y^2} = -\frac{x^2 + y^2}{y^3} \\
 &= -\frac{1}{y^3}
 \end{aligned}$$

所以 $f'(0) = 0, f''(0) = -1.$

前述引例: $e^{x+y} + xy = 0$

令 $F(x, y) = e^{x+y} + xy = 0$,

当 $F_y(x, y) = e^{x+y} + x \neq 0$ 时,

就可确定可导函数 $y = y(x)$, 且

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{e^{x+y} + y}{e^{x+y} + x}.$$

例 2 已知 $\ln \sqrt{x^2 + y^2} = \arctan \frac{y}{x}$, 求 $\frac{dy}{dx}$.

解 [法一]

$$\text{令 } F(x, y) = \ln \sqrt{x^2 + y^2} - \arctan \frac{y}{x},$$

$$\text{则 } F_x(x, y) = \frac{x + y}{x^2 + y^2}, \quad F_y(x, y) = \frac{y - x}{x^2 + y^2} \neq 0,$$

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{x + y}{y - x}.$$

[法二] 方程两边对 x 求导, 视 y 为 x 的函数:

2. 推广到三元以上 $F(x, y, z) = 0$

隐函数存在定理 2 设函数 $F(x, y, z)$ 在点 $P(x_0, y_0, z_0)$ 的某一邻域内有连续的偏导数, 且 $F(x_0, y_0, z_0) = 0$, $F_z(x_0, y_0, z_0) \neq 0$, 则方程 $F(x, y, z) = 0$ 在点 $P(x_0, y_0, z_0)$ 的某一邻域内恒能唯一确定一个单值连续且具有连续偏导数的函数 $z = f(x, y)$, 它满足条件 $z_0 = f(x_0, y_0)$,

并有
$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}.$$

例 3 设方程 $x^2 + y^2 + z^2 = 4$ 确定隐函数 $z = f(x, y)$,

求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, \frac{\partial^2 z}{\partial x^2}$ 和 $\frac{\partial^2 z}{\partial x \partial y}$.

(1) 先求一阶偏导数:

解法1 设 $F(x, y, z) = x^2 + y^2 + z^2 - 4z$,

则 $F_x = 2x, F_y = 2y, F_z = 2z - 4$.

于是 $\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{2x}{2z-4} = \frac{x}{2-z},$

$$\frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = -\frac{2y}{2z-4} = \frac{y}{2-z}.$$

解法2

方程 $x^2 + y^2 + z^2 = 4z$ 两端对 x 求偏导, 得

$$2x + 2z \frac{\partial z}{\partial x} = 4 \frac{\partial z}{\partial x},$$

解得

$$\frac{\partial z}{\partial x} = -\frac{2x}{2z-4} = \frac{x}{2-z},$$

同理方程两端对 y 求偏导, 得

$$\frac{\partial z}{\partial y} = -\frac{2y}{2z-4} = \frac{y}{2-z}.$$

解法3

方程 $x^2 + y^2 + z^2 = 4z$ 两端同时求微分，得

$$2xdx + 2ydy + 2zdz = 4dz,$$

整理得

$$dz = \frac{x}{2-z} dx + \frac{y}{2-z} dy,$$

所以

$$\frac{\partial z}{\partial x} = \frac{x}{2-z}, \quad \frac{\partial z}{\partial y} = \frac{y}{2-z}.$$

(2) 再求二阶偏导数:

解法1 $\frac{\partial z}{\partial x}$ 再对 x 或 y 求偏导数, 得

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \frac{2 - z + x \frac{\partial z}{\partial x}}{(2 - z)^2} = \frac{(2 - z)^2 + x^2}{(2 - z)^3},$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) = \frac{0 + x \frac{\partial z}{\partial y}}{(2 - z)^2} = \frac{xy}{(2 - z)^3}.$$

解法2 在第一步得到的方程 $x + z \frac{\partial z}{\partial x} = 2 \frac{\partial z}{\partial x}$

的两端再对 x 求偏导数，得

$$1 + \left(\frac{\partial z}{\partial x}\right)^2 + z \frac{\partial^2 z}{\partial x^2} = 2 \frac{\partial^2 z}{\partial x^2}$$

整理可得

$$\frac{\partial^2 z}{\partial x^2} = \frac{1 + \left(\frac{\partial z}{\partial x}\right)^2}{2 - z} = \frac{(2 - z)^2 + x^2}{(2 - z)^3},$$

同理，方程 $x + z \frac{\partial z}{\partial x} = 2 \frac{\partial z}{\partial x}$ 的两端再对 y

求偏导数，得

$$1 + \frac{\partial z}{\partial x} \cdot \frac{\partial z}{\partial y} + z \frac{\partial^2 z}{\partial x \partial y} = 2 \frac{\partial^2 z}{\partial x \partial y},$$

整理可得

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\frac{\partial z}{\partial x} \cdot \frac{\partial z}{\partial y}}{2 - z} = \frac{xy}{(2 - z)^3}.$$

例 4 设 $z = f(x, y)$ 是由方程 $z^3 + 2xz - xy = 1$ 确定的

隐函数, 求 $\left. \frac{\partial^2 z}{\partial x^2} \right|_{(0,1)}$.

解 由 $x = 0, y = 1$, 得 $z = 1$,

方程两边分别对 x, y 求偏导(视 $z = z(x, y)$), 得

$$3z^2 \cdot z_x + 2z + 2x \cdot z_x - y = 0 \quad (1)$$

将 $x = 0, y = 1, z = 1$, 代入 (1) 得

$$z_x \big|_{(0,1)} = -\frac{1}{3}$$

方程(1)两端再对 x 求偏导数, 得

$$6z \cdot (z_x)^2 + 3z^2 \cdot z_{xx} + 4z_x + 2x \cdot z_{xx} = 0 \quad (2)$$

将 $x = 0, y = 1, z = 1, z_x|_{(0,1)} = -\frac{1}{3}$ 代入(2), 得

$$z_{xx}|_{(0,1)} = \frac{2}{9}.$$

例 5 设 $z = f(x + y + z, xyz)$, 求 $\frac{\partial z}{\partial x}$, $\frac{\partial x}{\partial y}$, $\frac{\partial y}{\partial z}$.

解法1 设 $F(x, y, z) = f(x + y + z, xyz) - z$,

则 $F_x = f'_1 + yzf'_2$, $F_y = f'_1 + xzf'_2$,

$$F_z = f'_1 + xyf'_2 - 1,$$

于是
$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = \frac{f'_1 + yzf'_2}{1 - f'_1 - xyf'_2},$$

$$\frac{\partial x}{\partial y} = -\frac{F_y}{F_x} = -\frac{f'_1 + xzf'_2}{f'_1 + yzf'_2},$$

$$\frac{\partial y}{\partial z} = -\frac{F_z}{F_y} = \frac{1 - f'_1 - xyf'_2}{f'_1 + xzf'_2}.$$

解法2 把 z 看成 x, y 的函数对 x 求偏导数得

$$\frac{\partial z}{\partial x} = f_1' \cdot (1 + \frac{\partial z}{\partial x}) + f_2' \cdot (yz + xy \frac{\partial z}{\partial x}),$$

整理得
$$\frac{\partial z}{\partial x} = \frac{f_1' + yzf_2'}{1 - f_1' - xyf_2'},$$

把 x 看成 z, y 的函数对 y 求偏导数得

$$0 = f_1' \cdot (\frac{\partial x}{\partial y} + 1) + f_2' \cdot (xz + yz \frac{\partial x}{\partial y}),$$

整理得 $\frac{\partial x}{\partial y} = -\frac{f_1' + xzf_2'}{f_1' + yzf_2'}$,

把 y 看成 x, z 的函数对 z 求偏导数得

$$1 = f_1' \cdot \left(\frac{\partial y}{\partial z} + 1 \right) + f_2' \cdot (xy + xz \frac{\partial y}{\partial z}),$$

整理得 $\frac{\partial y}{\partial z} = \frac{1 - f_1' - xyf_2'}{f_1' + xzf_2'}.$

例 6 设 $F(u, v)$ 具有连续偏导数, 函数 $z = f(x, y)$ 由方程 $F(cx - az, cy - bz) = 0$ 确定. 证明 $a \frac{\partial z}{\partial x} + b \frac{\partial z}{\partial y} = c$.

解法1 设 $G(x, y, z) = F(cx - az, cy - bz)$,

$$\text{则 } G_x = cF'_1, \quad G_y = cF'_2, \quad G_z = -aF'_1 - bF'_2,$$

$$\text{于是 } \frac{\partial z}{\partial x} = -\frac{G_x}{G_z} = \frac{cF'_1}{aF'_1 + bF'_2},$$

$$\frac{\partial z}{\partial y} = -\frac{G_y}{G_z} = \frac{cF'_2}{aF'_1 + bF'_2},$$

$$\text{所以 } a \frac{\partial z}{\partial x} + b \frac{\partial z}{\partial y} = \frac{acF'_1 + bcF'_2}{aF'_1 + bF'_2} = c.$$

解法2

方程 $F(cx - az, cy - bz) = 0$ 两端对 x 求偏导, 得

$$F_1' \cdot (c - a \frac{\partial z}{\partial x}) + F_2' \cdot (-b \frac{\partial z}{\partial x}) = 0,$$

解得

$$\frac{\partial z}{\partial x} = \frac{cF_1'}{aF_1' + bF_2'},$$

同理方程两端对 y 求偏导, 得

$$\frac{\partial z}{\partial y} = \frac{cF_2'}{aF_1' + bF_2'},$$

所以

$$a \frac{\partial z}{\partial x} + b \frac{\partial z}{\partial y} = \frac{acF_1' + bcF_2'}{aF_1' + bF_2'} = c.$$

解法3

方程 $F(cx - az, cy - bz) = 0$ 两端同时求微分, 得

$$F_1' \cdot (cdx - adz) + F_2' \cdot (cdy - adz) = 0,$$

整理得

$$dz = \frac{cF_1'}{aF_1' + bF_2'} dx + \frac{cF_2'}{aF_1' + bF_2'} dy,$$

从而

$$\frac{\partial z}{\partial x} = \frac{cF_1'}{aF_1' + bF_2'}, \quad \frac{\partial z}{\partial y} = \frac{cF_2'}{aF_1' + bF_2'},$$

所以

$$a \frac{\partial z}{\partial x} + b \frac{\partial z}{\partial y} = \frac{acF_1' + bcF_2'}{aF_1' + bF_2'} = c.$$

二、方程组的情形 (1) $\begin{cases} F(x, y, z) = 0 \\ G(x, y, z) = 0 \end{cases}$

隐函数存在定理 3 设 $F(x, y, z)$ 、 $G(x, y, z)$ 在点 $P(x_0, y_0, z_0)$ 的某一邻域内有对各个变量的连续偏导数，且 $F(x_0, y_0, z_0) = 0$ ， $G(x_0, y_0, z_0) = 0$ ，且偏导数所组成的函数行列式（或称雅可比式）

$$J = \frac{\partial(F, G)}{\partial(y, z)} = \begin{vmatrix} \frac{\partial F}{\partial y} & \frac{\partial F}{\partial z} \\ \frac{\partial G}{\partial y} & \frac{\partial G}{\partial z} \end{vmatrix} \quad \text{在点 } P(x_0, y_0, z_0) \text{ 不等于零,}$$

则方程组 $\begin{cases} F(x, y, z) = 0 \\ G(x, y, z) = 0 \end{cases}$ 在点 $P(x_0, y_0, z_0)$ 的某一邻域内恒能唯一确定一对单值连续且具有连续偏导数的函数 $\begin{cases} y = y(x) \\ z = z(x) \end{cases}$, , 它们满足条件 $y_0 = y(x_0), z_0 = z(x_0)$, 并有

$$\frac{dy}{dx} = -\frac{\frac{\partial(F, G)}{\partial(x, z)}}{\frac{\partial(F, G)}{\partial(y, z)}} = -\frac{\begin{vmatrix} F_x & F_z \\ G_x & G_z \end{vmatrix}}{\begin{vmatrix} F_y & F_z \\ G_y & G_z \end{vmatrix}}, \quad \frac{dz}{dx} = -\frac{\frac{\partial(F, G)}{\partial(y, x)}}{\frac{\partial(F, G)}{\partial(y, z)}} = -\frac{\begin{vmatrix} F_y & F_x \\ G_y & G_x \end{vmatrix}}{\begin{vmatrix} F_y & F_z \\ G_y & G_z \end{vmatrix}},$$

例 1 设 $\begin{cases} x^2 + y^2 + z^2 = 6 \\ x + y + z = 0 \end{cases}$ 确定 $y = y(x)$, $z = z(x)$,

求 $\frac{dy}{dx}$, $\frac{dz}{dx}$.

解1 直接代入公式.

解2 运用公式推导的方法.

将所给方程的两边分别对 x 求导,
视 $y = y(x)$, $z = z(x)$.

$$(2) \quad \begin{cases} F(x, y, u, v) = 0 \\ G(x, y, u, v) = 0 \end{cases}$$

隐函数存在定理 4 设 $F(x, y, u, v)$ 、 $G(x, y, u, v)$ 在点 $P(x_0, y_0, u_0, v_0)$ 的某一邻域内有对各个变量的连续偏导数，且 $F(x_0, y_0, u_0, v_0) = 0$ ， $G(x_0, y_0, u_0, v_0) = 0$ ，且偏导数所组成的函数行列式（或称雅可比式）

$$J = \frac{\partial(F, G)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial F}{\partial u} & \frac{\partial F}{\partial v} \\ \frac{\partial G}{\partial u} & \frac{\partial G}{\partial v} \end{vmatrix}$$

在点 $P(x_0, y_0, u_0, v_0)$ 不等于零, 则方程组

$$F(x, y, u, v) = 0, \quad G(x, y, u, v) = 0$$

在点 $P(x_0, y_0, u_0, v_0)$ 的某一邻域内恒能唯一确定一组单值连续且具有连续偏导数的函数 $u = u(x, y)$, $v = v(x, y)$, 它们满足条件 $u_0 = u(x_0, y_0)$, $v_0 = v(x_0, y_0)$, 并有

$$\frac{\partial u}{\partial x} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(x, v)} = -\frac{\begin{vmatrix} F_x & F_v \\ G_x & G_v \end{vmatrix}}{\begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}},$$

$$\frac{\partial v}{\partial x} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(u, x)} = -\frac{\begin{vmatrix} F_u & F_x \\ G_u & G_x \end{vmatrix}}{\begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}}$$

$$\frac{\partial u}{\partial y} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(y, v)} = -\frac{\begin{vmatrix} F_y & F_v \\ G_y & G_v \end{vmatrix}}{\begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}},$$

$$\frac{\partial v}{\partial y} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(u, y)} = -\frac{\begin{vmatrix} F_u & F_y \\ G_u & G_y \end{vmatrix}}{\begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}}.$$

例2 设 $xu - yv = 0$, $yu + xv = 1$,

求 $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$ 和 $\frac{\partial v}{\partial y}$.

解1 直接代入公式;

解2 运用公式推导的方法,

将所给方程的两边对 x 求偏导并移项

$$\begin{cases} x \frac{\partial u}{\partial x} - y \frac{\partial v}{\partial x} = -u \\ y \frac{\partial u}{\partial x} + x \frac{\partial v}{\partial x} = -v \end{cases}, \quad J = \begin{vmatrix} x & -y \\ y & x \end{vmatrix} = x^2 + y^2,$$

在 $J \neq 0$ 的条件下,

$$\frac{\partial u}{\partial x} = \frac{\begin{vmatrix} -u & -y \\ -v & x \end{vmatrix}}{\begin{vmatrix} x & -y \\ y & x \end{vmatrix}} = -\frac{xu + yv}{x^2 + y^2}, \quad \frac{\partial v}{\partial x} = \frac{\begin{vmatrix} x & -u \\ y & -v \end{vmatrix}}{\begin{vmatrix} x & -y \\ y & x \end{vmatrix}} = \frac{yu - xv}{x^2 + y^2},$$

将所给方程的两边对 y 求导, 用同样方法得

$$\frac{\partial u}{\partial y} = \frac{xv - yu}{x^2 + y^2}, \quad \frac{\partial v}{\partial y} = -\frac{xu + yv}{x^2 + y^2}.$$

例1: 设 $z = \tan(x + y)$, y 由方程 $e^{x-y} = xy$ 确定, 求 $\frac{dz}{dx}$.

解 $\frac{dz}{dx} = \sec^2(x + y)(1 + y')$ 可求全导数.

$$\text{令 } F(x, y) = e^{x-y} - xy = 0,$$

$$F_x = e^{x-y} - y, \quad F_y = -e^{x-y} - x,$$

$$y' = -\frac{F_x}{F_y} = \frac{e^{x-y} - y}{e^{x-y} + x},$$

$$\frac{dz}{dx} = \sec^2(x + y)(1 + y') = \sec^2(x + y)\left(1 + \frac{e^{x-y} - y}{e^{x-y} + x}\right).$$

例2: 设 $z = z(x, y)$ 是由方程 $z^5 - xz^4 + yz^3 = 1$ 确定的

隐函数, 求 $\left. \frac{\partial^2 z}{\partial x \partial y} \right|_{\substack{x=0 \\ y=0}}$

解: 由 $x = 0, y = 0$, 得 $z = 1$,

方程两边分别对 x, y 求偏导(视 $z = z(x, y)$), 得

$$5z^4 \cdot z_x - z^4 - 4xz^3 \cdot z_x + 3yz^2 \cdot z_x = 0 \quad (1)$$

$$5z^4 \cdot z_y - 4xz^3 \cdot z_y + z^3 + 3yz^2 \cdot z_y = 0 \quad (2)$$

将 $x = 0, y = 0, z = 1$ 代入(1)、(2)得:

$$z_x|_{(0,0)} = \frac{1}{5}, \quad z_y|_{(0,0)} = -\frac{1}{5},$$

方程(1)两边对 y 求偏导(z 、 z_x 均为 x 、 y 的函数):

$$2z(10z^2 - 6xz + 3y) \cdot z_x \cdot z_y + z^2(5z^2 - 4xz + 3y) \cdot z_{xy} - 4z^3 \cdot z_y + 3z^2 \cdot z_x = 0 \quad (3)$$

将 $x = 0, y = 0, z = 1, z_x|_{(0,0)} = \frac{1}{5}, z_y|_{(0,0)} = -\frac{1}{5}$ 代入(3)得:

$$z_{xy}|_{(0,0)} = -\frac{3}{25}.$$

例3: 已知 $F(x, x+y, x+y+z)=0$, 求 $\frac{\partial z}{\partial x}$.

解: 方程两边对 x 求偏导

$$F_1' + F_2' + F_3'(1 + \frac{\partial z}{\partial x}) = 0$$

$$\Rightarrow \frac{\partial z}{\partial x} = -\frac{F_1' + F_2' + F_3'}{F_3'}$$

例4: $f(x, y, z) = xy^2z^3$, 而 x, y, z 满足

$$x^2 + y^2 + z^2 - 3xyz = 0, \text{ 求 } \frac{\partial f}{\partial x}$$

解: 由方程可确定 $z = z(x, y)$, $\frac{\partial f}{\partial x} = y^2(z^3 + 3xz^2 \frac{\partial z}{\partial x})$

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{2x - 3yz}{2z - 3xy},$$

$$\Rightarrow \frac{\partial f}{\partial x} = y^2z^3 - 3xy^2z^2 \cdot \frac{2x - 3yz}{2z - 3xy}$$

四、小结

隐函数的求导法则（分以下几种情况）

$$(1) \quad F(x, y) = 0 \quad (2) \quad F(x, y, z) = 0$$

$$(3) \quad \begin{cases} F(x, y, z) = 0 \\ G(x, y, z) = 0 \end{cases} \quad (4) \quad \begin{cases} F(x, y, u, v) = 0 \\ G(x, y, u, v) = 0 \end{cases}$$

思考题

已知 $\frac{x}{z} = \varphi\left(\frac{y}{z}\right)$, 其中 φ 为可微函数,

$$\text{求 } x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = ?$$

思考题解答

$$\text{记 } F(x, y, z) = \frac{x}{z} - \varphi\left(\frac{y}{z}\right), \quad \text{则 } F_x = \frac{1}{z},$$

$$F_y = -\varphi'\left(\frac{y}{z}\right) \cdot \frac{1}{z}, \quad F_z = \frac{-x}{z^2} - \varphi'\left(\frac{y}{z}\right) \cdot \frac{(-y)}{z^2},$$

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = \frac{z}{x - y\varphi'\left(\frac{y}{z}\right)}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = \frac{-z\varphi'\left(\frac{y}{z}\right)}{x - y\varphi'\left(\frac{y}{z}\right)},$$

$$\text{于是 } x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = z.$$

练习题

一、填空题:

1、设 $\ln \sqrt{x^2 + y^2} = \arctan \frac{y}{x}$, 则

$$\frac{dy}{dx} = \underline{\hspace{10cm}}.$$

2、设 $z^x = y^z$, 则

$$\frac{\partial z}{\partial x} = \underline{\hspace{10cm}},$$

$$\frac{\partial z}{\partial y} = \underline{\hspace{10cm}}.$$

二、设 $2\sin(x + 2y - 3z) = x + 2y - 3z$,

证明: $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 1.$

三、如果函数 $f(x, y, z)$ 对任何 t 恒满足关系式 $f(tx, ty, tz) = t^k f(x, y, z)$, 则称函数 $f(x, y, z)$ 为 k 次齐次函数, 试证: k 次齐次函数满足方程

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + z \frac{\partial f}{\partial z} = kf(x, y, z).$$

四、设 $z^3 - 3xyz = a^3$, 求 $\frac{\partial^2 z}{\partial x \partial y}$.

五、求由下列方程组所确定的函数的导数或偏导数:

1、设 $\begin{cases} z = x^2 + y^2 \\ x^2 + 2y^2 + 3z^2 = 20 \end{cases}$, 求 $\frac{dy}{dx}, \frac{dz}{dx}$.

2、设 $\begin{cases} u = f(ux, v + y) \\ v = g(u - x, v^2 y) \end{cases}$, 求 $\frac{\partial u}{\partial x}, \frac{\partial v}{\partial x}$.

(其中 f, g 具有一阶连续偏导数)

六、设函数 $u(x)$ 由方程组
$$\begin{cases} u = f(x, y) \\ g(x, y, z) = 0 \\ h(x, z) = 0 \end{cases}$$
 所确定,

且 $\frac{\partial g}{\partial y} \neq 0, \frac{\partial h}{\partial z} \neq 0$, 求 $\frac{du}{dx}$. (f, g, h 均可微)

七、设 $y = f(x, t)$, 而 t 是由方程 $F(x, y, t) = 0$ 所确定的 x, y 的函数, 求 $\frac{dy}{dx}$.

八、设 $z = z(x, y)$ 由方程 $F(x + \frac{x}{y}, y + \frac{z}{x}) = 0$ 所确定,

证明: $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = z - xy$.

练习题答案

$$\text{一、 } 1、\frac{x+y}{x-y}; \quad 2、\frac{-z^x \ln z}{xz^{x-1} - y^z \ln y};$$

$$3、\frac{zy^{z-1}}{xz^{x-1} - y^z \ln y}.$$

$$\text{四、 } \frac{\partial^2 z}{\partial x \partial y} = \frac{z(z^4 - 2xyz^2 - x^2 y^2)}{(z^2 - xy)^3}.$$

$$\text{五、 } 1、\frac{dy}{dx} = \frac{-x(6z+1)}{2y(3z+1)}, \frac{dz}{dx} = \frac{x}{3z+1};$$

$$2、\frac{\partial u}{\partial x} = \frac{-uf'_1(2yvg'_2 - 1) - f'_2 \cdot g'_1}{(xf'_1 - 1)(2yvg'_2 - 1) - f'_2 \cdot g'_1},$$

$$\frac{\partial v}{\partial x} = \frac{g'_1(xf'_1 + uf'_1 - 1)}{(xf'_1 - 1)(2yvg'_2 - 1) - f'_2 \cdot g'_1}.$$

$$\begin{aligned}
 \text{六、} \quad \frac{du}{dx} &= f'_x - \frac{f'_x \cdot g'_x}{g'_y} + \frac{f'_y \cdot g'_z \cdot h'_x}{g'_y \cdot h'_z} \\
 &= \frac{f'_x g'_y h'_z - f'_x g'_x h'_z + f'_y g'_z h'_x}{g'_y h'_z}.
 \end{aligned}$$

$$\text{七、} \quad \frac{dy}{dx} = \frac{F'_t \cdot f'_x - F'_x \cdot f'_t}{F'_t + F'_y \cdot f'_t}.$$